

Accurate multislice theory for elastic electron scattering in transmission electron microscopy

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Abstract

切片传输算子矩阵

传递矩阵技术

Combining the concept of the slice-transmission-operator matrix with the transfer matrix technique, the complete Schrödinger equation for elastic electron scattering in solids is rigorously solved within the framework of the multislice scheme. This new approach extends the range in which the multislice theory is accurate. It is shown that the multislice theory can in principle be as accurate as the Bloch wave theory and the Green function theory in all cases of elastic electron scattering. It is pointed out that in some cases such as the calculation of high-order Laue zone (HOLZ) reflections, more accurate corrections should be made for the conventional high-energy approximation and therefore the conventional multislice method (CMS). A new multislice formula is developed for calculating the transmitted wave. This formula takes back-scattering effects into account and may therefore be accurate for large beam tilt. It is also shown that the new multislice approach keeps all the advantages of the CMS with respect to the Bloch wave method.

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Keywords: Multislice theory; HOLZ effects; Back-scattering effects

1. Introduction

电子能量过滤器 (electron energy-filters)

高相干场发射电子源

Recent developments of high-energy electron microscopes (HEEMs), where highly coherent field-emission electron sources, electron energy-filters and slow scan CCD (charge-coupled device) cameras are becoming an integrated part of the instruments, have made it possible to obtain reliable quantitative experimental data with a large flexibility in the choice of the experiment (HREM/STEM/focal images, ED, CBED, ptychography, energy selection, etc.). However, the interpretation of the experimental data requires that also the theories are sufficiently accurate so as to enable quantitative matching with the experiment. For example,

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慢扫描CCD (slow scan charge-couple device) (电荷耦合设备)

experimental diffraction patterns or images digitized by the CCD camera into the computer are quantitatively compared with theoretically calculated patterns or images [1, 2]. Although many excellent theories and simulation methods have so far been established for the analysis and calculation of high-energy electron diffraction (HEED) and imaging [2], the current situation is still such that information available in modern HEEDs is not adequately exploited.

For quantitative electron microscopy, multiple elastic electron scattering is the dominant interaction process by which the electron contains the structural information about the object. Accurate simulations of high-energy elastic electron scattering involve the use of many-beam dynamical diffraction theories. Currently there are two popular theories which are mostly used for many-beam dynamical calculations [1], i.e., the Bloch wave theory and the multislice theory. Other theories exist which are convenient for particular cases of HEED [2]. Among all the existing theories, the Green function theory is in principle the most accurate since it gives the most general solution for the complete Schrödinger-like equation for high-energy elastic electron scattering in solids without any assumption for approximations. But it is less useful for calculating HEED and imaging due to the slow convergence of the Born series [2]. In addition, Van Dyck [3] also showed that using the scattering matrix method, the complete Schrödinger equation can rigorously be solved but the obtained solution is too complicated for numerical evaluations.

The Bloch wave theory (sometimes referred to the Bethe theory [4]), which includes the reciprocal-space method [2, 4], and the real-space method (see Ref. [5] for a review), has been the most fundamental theory for understanding and calculating the dynamical diffraction effects in TEM [6, 7]. Although most of the Bloch wave procedures are developed for calculating only the forward-scattered wave, the back-scattered wave can in principle also be included in the Bloch wave method [8]. However, this theory is not convenient for defect structures [2, 8] and cannot include top and bottom surface effects [9] since it is based on the assumption of a perfectly periodic potential. It is less efficient than the multislice method in computations when a large number of beams are involved, e.g., in the case of HRTEM image simulations [10, 11].

The multislice method, originally proposed by Cowley and Moodie [12] from physico-optics principles, has been the most efficient procedure in many cases of the dynamical calculation for HEED and imaging [1, 2], especially when it is speeded up by the FFT technique [13]. Another important advantage of the multislice method over the Bloch wave method is that it does not require the periodicity of the potential and can therefore be used for aperiodic structures [9, 14]. In addition, the multislice method can also be used for the calculation of inelastic scattering effects [15, 16]. It is shown [17] that all the multislice methods can be studied in a general framework as integration methods for the reduced or modified Schrödinger equation for HEED. However, this equation is based on the high-energy approximation, in which back-scattering effects are neglected. Hence in case these effects are not negligible, the current multislice methods will not be valid.

It is known that none of the two most popular theories is generally valid for all the cases of HEED. In fact, both of the two methods have shown their difficulties in dealing with RHEED patterns and images, so that other approaches have to be developed for this type of dynamical calculations (for a review Ref. [2]). For instance, the method, originally proposed by Lynch and Moodie [18] for calculating low-energy electron diffraction (LEED), has successfully been applied to RHEED [19, 20]. Generally, when the incident beam is largely tilted away from the surface normal, the conventional high-energy approximation will not be valid and back-scattering effects have to be taken into account [5, 8]. RHEED is only the extreme case where back-scattering effects are dominant. Hence, as more and more advanced experimental techniques become available, there is an urgent need for more accurate but efficient dynamical theories (or simulation methods).

The purpose of this work is to extend the validity of the multislice theory or method, so that it accounts for the back-scattering effects and at the same time keeps all the advantages of the conventional multislice concept.

2. General solution of the Schrödinger equation in the multislice scheme

To include back-scattering effects of a high-energy electron wave passing a solid film (Fig. 1a), we have to solve the “relativistically corrected” Schrödinger-like equation, which is derived from the Dirac equation for HEED [21–25]

$$\left[\frac{h^2}{8\pi^2 m} \nabla_r^2 + eU(\mathbf{r}) + \frac{h^2 K_0^2}{2m} \right] \Psi(\mathbf{r}) = 0, \quad (1)$$

where $U(\mathbf{r})$ is the crystal potential, $\Psi(\mathbf{r})$ is the wave function of the high-energy electron, ∇_r^2 is the three-dimensional (3D) Laplacian operator, e is the electron charge, h is the Planck constant, m and K_0 ($= 1/\lambda$, λ is the wavelength), respectively, are the “relativistically corrected” mass and wave vector of the incident electron.

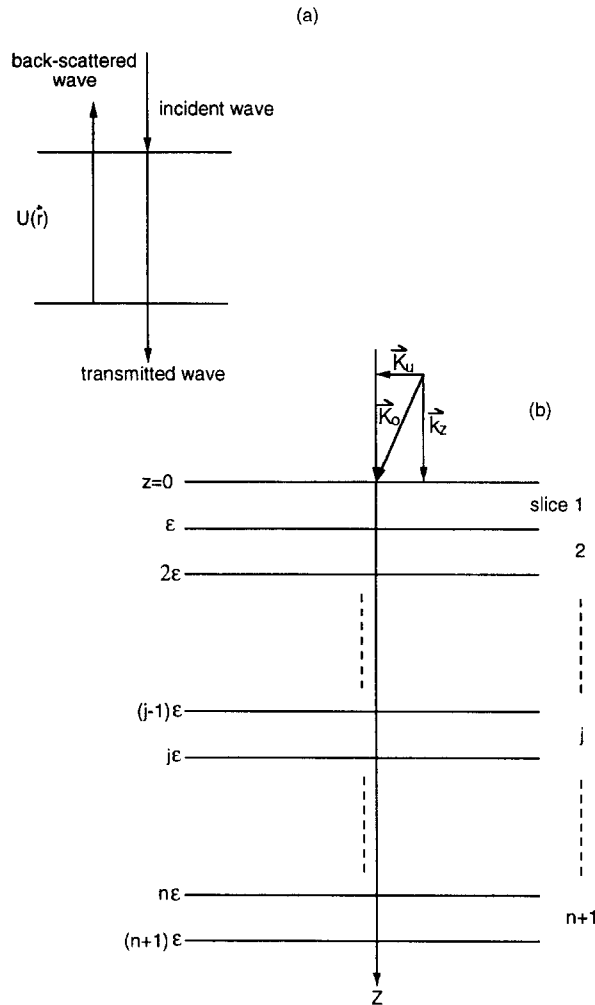


Fig. 1. Schematic illustration of the elastic electron scattering in a solid film: (a) forward-scattering and back-scattering; (b) multislice scheme.

We now try to solve Eq. (1) within the framework of the multislice scheme, in which the solid is considered as a combination of a series of very thin slices parallel to the surface (Fig. 1b) and the potential of each slice is averaged along the z -axis:

$$\bar{U}_j(\mathbf{R}) = \frac{1}{\varepsilon} \int_{(j-1)\varepsilon}^{j\varepsilon} U(\mathbf{r}) dz,$$

where $\mathbf{R} = (x, y)$ and ε is the slice thickness. Without any loss of generality, Eq. (1) can be rewritten as

$$\frac{\partial^2 \Psi(\mathbf{r})}{\partial z^2} + [2\pi \hat{k}(\mathbf{r})]^2 \Psi(\mathbf{r}) = 0, \quad (2)$$

where wave-vector operator $\hat{k}(\mathbf{r})$ is defined by

$$\begin{aligned} \hat{k}(\mathbf{r}) &= \pm K_0 \sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta + \frac{\sigma}{\pi K_0} U(\mathbf{r})} \\ &= \pm K_0 \left[1 + \frac{1}{2\pi K_0} \left(\frac{\Delta}{4\pi K_0} + \sigma U \right) - \frac{1}{8(\pi K_0)^2} \left(\frac{\Delta}{4\pi K_0} + \sigma U \right)^2 + \dots \right] \end{aligned} \quad (3)$$

with $\Delta = (\partial^2/\partial x^2) + (\partial^2/\partial y^2)$, the 2D Laplacian operator and $\sigma = 2\pi e\lambda m/h^2$, the interaction constant.

To our knowledge, for the whole solid film it is impossible to find a simple solution for Eq. (2). But for a single slice, e.g., slice j , we have

$$\frac{\partial^2 \Psi^j(\mathbf{r})}{\partial z^2} + (2\pi \hat{k}_j)^2 \Psi^j(\mathbf{r}) = 0, \quad (4)$$

where $j = 1, 2, \dots, n+1$, and $\hat{k}_j$ is independent of z with

$$\hat{k}_j(\mathbf{R}) = \pm K_0 \sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta + \frac{\sigma}{\pi K_0} U_j(\mathbf{R})}. \quad (5)$$

So the wave function within slice j (from $z = (j-1)\varepsilon$ to $z = j\varepsilon$) is found as

$$\Psi^j(\mathbf{R}, z) = e^{2\pi i \hat{k}_j z'} \Phi_j^f[\mathbf{R}, (j-1)\varepsilon] + e^{-2\pi i \hat{k}_j z'} \Phi_j^b[\mathbf{R}, (j-1)\varepsilon], \quad (6)$$

where $z'_j = z - (j-1)\varepsilon$, Φ_j^f and Φ_j^b , respectively, indicate the forward-scattered wave and the back-scattered wave at $z = (j-1)\varepsilon$. It should be noted that since $\hat{k}_j$ is an operator acting on (x, y) , Eq. (6) is the only correct form of the general solution for Eq. (4). For example, $\Phi_j^f(\mathbf{R})e^{2\pi i \hat{k}_j z'_j}$ will not satisfy Eq. (4).

For the entire problem, what we know are the potential of the solid and the boundary conditions:

(i) at the entrance surface: $\Phi_1^f(\mathbf{R}, z=0) = e^{2\pi i \mathbf{K}_u \cdot \mathbf{R}}$, where $\mathbf{K}_u$ indicates the inclined illumination [26] (for the normal illumination $\mathbf{K}_u = 0$ and therefore $\Phi_1^f = 1$);

(ii) at the exit surface: $\Phi_{n+1}^b(\mathbf{R}, z=n\varepsilon) = 0$.

What we finally want to know are the forward-scattered wave at $z = n\varepsilon$ and the back-scattered wave at $z = 0$, i.e., $\Phi_{n+1}^f(\mathbf{R}, z=n\varepsilon)$ and $\Phi_1^b(\mathbf{R}, z=0)$. It should be stressed that in the multislice scheme, in order to use the boundary conditions and include the surface effects correctly [27], it is necessary to include some vacuum slices above the top surface and below the bottom surface so that we should have at least $\bar{U}_1(\mathbf{R}) = 0$, $\bar{U}_n(\mathbf{R}) = 0$ and $\bar{U}_{n+1}(\mathbf{R}) = 0$.

Since the wavefunctions should smoothly be connected at the interfaces between two successive slices, at $z = (j-1)\varepsilon$, we have

$$\Psi^{j-1}|_{z=(j-1)\varepsilon} = \Psi^j|_{z=(j-1)\varepsilon} \quad (7)$$

and

$$\left. \frac{\partial \Psi^{j-1}}{\partial z} \right|_{z=(j-1)\epsilon} = \left. \frac{\partial \Psi^j}{\partial z} \right|_{z=(j-1)\epsilon}, \quad (8)$$

that is,

$$e^{2\pi i \hat{k}_{j-1} \epsilon} \Phi_{j-1}^f + e^{-2\pi i \hat{k}_{j-1} \epsilon} \Phi_{j-1}^b = \Phi_j^f + \Phi_j^b \quad (9)$$

and

$$\hat{k}_{j-1}(e^{2\pi i \hat{k}_{j-1} \epsilon} \Phi_{j-1}^f - e^{-2\pi i \hat{k}_{j-1} \epsilon} \Phi_{j-1}^b) = \hat{k}_j(\Phi_j^f - \Phi_j^b). \quad (10)$$

Since there are n interfaces between $n+1$ slices, from Eqs. (9) and (10) we have $2n$ equations for $2(n+1)$ variables. But two of them, i.e., Φ_1^f and Φ_{n+1}^b , are known from the boundary conditions, so the problem is solvable. On the other hand, for our purpose it is not necessary to know the intermedium variables. What we should do is to connect Φ_{n+1}^f and Φ_1^b to Φ_1^f and Φ_{n+1}^b . Using the so-called transfer matrix technique (e.g., Ref. [28]), this can be done in a very concise way:

(1) Using a 2×2 matrix of operators, Eqs. (9) and (10) can be rewritten as

$$\begin{pmatrix} \Phi_j^f \\ \Phi_j^b \end{pmatrix} = \hat{S}_{j-1} \begin{pmatrix} \Phi_{j-1}^f \\ \Phi_{j-1}^b \end{pmatrix} = \begin{pmatrix} \hat{S}_{j-1}^{11} & \hat{S}_{j-1}^{12} \\ \hat{S}_{j-1}^{21} & \hat{S}_{j-1}^{22} \end{pmatrix} \begin{pmatrix} \Phi_{j-1}^f \\ \Phi_{j-1}^b \end{pmatrix}, \quad (11)$$

where the slice-transmission-operator (STO) matrix is defined as follows:

$$\begin{pmatrix} \hat{S}_{j-1}^{11} & \hat{S}_{j-1}^{12} \\ \hat{S}_{j-1}^{21} & \hat{S}_{j-1}^{22} \end{pmatrix} = \begin{pmatrix} \hat{F}_{j,j-1} e^{2\pi i \hat{k}_{j-1} \epsilon} & \hat{B}_{j,j-1} e^{-2\pi i \hat{k}_{j-1} \epsilon} \\ \hat{B}_{j,j-1} e^{2\pi i \hat{k}_{j-1} \epsilon} & \hat{F}_{j,j-1} e^{-2\pi i \hat{k}_{j-1} \epsilon} \end{pmatrix} \quad (12)$$

with forward-scattering-coefficient (FSC) operator $\hat{F}_{j,j-1}$ and back-scattering-coefficient (BSC) operator $\hat{B}_{j,j-1}$:

$$\begin{aligned} \hat{F}_{j,j-1} &= \frac{\hat{k}_j + \hat{k}_{j-1}}{2\hat{k}_j} = 1 - \hat{B}_{j,j-1}, \\ \hat{B}_{j,j-1} &= \frac{\hat{k}_j - \hat{k}_{j-1}}{2\hat{k}_j}. \end{aligned} \quad (13)$$

(2) From Eq. (11) we obtain

$$\begin{pmatrix} \Phi_{n+1}^f \\ \Phi_{n+1}^b \end{pmatrix} = \hat{S}_n \cdots \hat{S}_1 \begin{pmatrix} \Phi_1^f \\ \Phi_1^b \end{pmatrix} = \hat{S}^n \begin{pmatrix} \Phi_1^f \\ \Phi_1^b \end{pmatrix} \quad (14)$$

with the multislice-transmission-operator (MSTO) matrix $\hat{S}^n$ defined by

$$\hat{S}^n = \begin{pmatrix} \hat{S}_n^{11} & \hat{S}_n^{12} \\ \hat{S}_n^{21} & \hat{S}_n^{22} \end{pmatrix} = \begin{pmatrix} \hat{S}_n^{11} & \hat{S}_n^{12} \\ \hat{S}_n^{21} & \hat{S}_n^{22} \end{pmatrix} \begin{pmatrix} \hat{S}_{n-1}^{11} & \hat{S}_{n-1}^{12} \\ \hat{S}_{n-1}^{21} & \hat{S}_{n-1}^{22} \end{pmatrix} \cdots \begin{pmatrix} \hat{S}_1^{11} & \hat{S}_1^{12} \\ \hat{S}_1^{21} & \hat{S}_1^{22} \end{pmatrix}. \quad (15)$$

(3) From Eqs. (14) and (15), we find out

$$\begin{pmatrix} \Phi_{n+1}^f \\ \Phi_{n+1}^b \end{pmatrix} = \begin{pmatrix} \hat{S}_{11}^n \Phi_1^f + \hat{S}_{12}^n \Phi_1^b \\ \hat{S}_{21}^n \Phi_1^f + \hat{S}_{22}^n \Phi_1^b \end{pmatrix}, \quad (16)$$

that is,

$$\Phi_1^b = (\hat{s}_{22}^n)^{-1} (\Phi_{n+1}^b - \hat{s}_{21}^n \Phi_1^f), \quad (17)$$

$$\Phi_{n+1}^f = \hat{s}_{11}^n \Phi_1^f + \hat{s}_{12}^n (\hat{s}_{22}^n)^{-1} (\Phi_{n+1}^b - \hat{s}_{21}^n \Phi_1^f), \quad (18)$$

where $(\hat{s}_{22}^n)^{-1}$ is defined by $(\hat{s}_{22}^n)^{-1} \hat{s}_{22}^n = \hat{s}_{22}^n (\hat{s}_{22}^n)^{-1} = 1$.

Putting $\Phi_{n+1}^b = 0$, we have

$$\Phi_1^b = -(\hat{s}_{22}^n)^{-1} \hat{s}_{21}^n \Phi_1^f, \quad (19)$$

$$\Phi_{n+1}^f = (\hat{s}_{11}^n - \hat{s}_{12}^n (\hat{s}_{22}^n)^{-1} \hat{s}_{21}^n) \Phi_1^f. \quad (20)$$

So the complete Schrödinger equation is rigorously solved in the multislice scheme. Once we know the potential of the solid film and the incident wavefunction, the back-scattered wave at the top-surface and the transmitted-wave at the bottom surface, respectively, can in principle be calculated from Eqs. (19) and (20). However, for calculation purposes we need more practical formulations. This will be done in Section 4.

3. Correction for the conventional forward-scattering approximation

Although it is still difficult to calculate, the rigorous multislice solution which includes the back-scattering effects can immediately yield a relatively simple solution for the forward-scattering wave, which is more accurate than the conventional forward-scattering approximation (it is often called the high-energy approximation) [29, 30].

3.1. Reduction of the schrödinger equation for forward scattering

Supposing the back-scattering effects are negligible, i.e., setting FSC operators $\hat{F}_{j,j-1} = 1$ and BSC operators $\hat{B}_{j,j-1} = 0$, through Eqs. (11)–(20) we have the following solution: $\Phi_1^b = 0$ and

$$\Phi_{n+1}^f(\mathbf{R}) = \left(\prod_{j=1}^n e^{2\pi i \hat{k}_j(\mathbf{R})\epsilon} \right) \Phi_1^f(\mathbf{R}). \quad (21)$$

It is interesting to notice that Eq. (21) is a typical multislice formula (with slice-transmission operator $e^{2\pi i \hat{k}_j(\mathbf{R})\epsilon}$) and when the slice thickness approaches zero it is the exact solution of the following reduced Schrödinger equation:

$$\frac{\partial \Phi(\mathbf{R}, z)}{\partial z} = 2\pi i \hat{k}_j(\mathbf{R}, z) \Phi(\mathbf{R}, z) = 2\pi i K_0 \sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta + \frac{\sigma}{\pi K_0} U(\mathbf{R}, z)} \Phi(\mathbf{R}, z). \quad (22)$$

Setting $\Phi(\mathbf{R}, z) = \varphi(\mathbf{R}, z) e^{2\pi i K_0 z}$ (without any loss of generality) and substituting it into Eq. (22), we have

$$\begin{aligned} \frac{\partial \varphi(\mathbf{R}, z)}{\partial z} &= 2\pi i K_0 \left[\sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta + \frac{\sigma}{\pi K_0} U(\mathbf{R}, z)} - 1 \right] \varphi(\mathbf{R}, z) \\ &= \left\{ i \left[\frac{\Delta}{4\pi K_0} + \sigma U(\mathbf{R}, z) \right] - \frac{i}{4\pi K_0} \left[\frac{\Delta}{4\pi K_0} + \sigma U(\mathbf{R}, z) \right]^2 + \dots \right\} \varphi(\mathbf{R}, z). \end{aligned} \quad (23)$$

So we see that (i). Neglecting the back-scattering effects, the Schrödinger equation can be reduced to a first-order differential equation in z , (ii). The conventional high-energy approximation for

forward-scattering is only a first-order approximation of Eq. (23) in the electron wavelength, which is [5, 17]

$$\frac{\partial \varphi(\mathbf{R}, z)}{\partial z} = i \left[\frac{\Delta}{4\pi K_0} + \sigma U(\mathbf{R}, z) \right] \varphi(\mathbf{R}, z). \quad (24)$$

It has been noticed that using the parabolic approximation for the excitation error, Eq. (24) may cause observable errors in the calculation of HOLZ patterns and therefore in the real-space Bloch wave (perturbation) method and the CMS based on Eq. (24), one should use a corrected form for the “parabola” excitation error [5, 31, 32]. This correction can easily be included in Eq. (24) so that

$$\begin{aligned} \frac{\partial \varphi(\mathbf{R}, z)}{\partial z} &= i \left\{ 2\pi K_0 \left[\sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta} - 1 \right] + \sigma U(\mathbf{R}, z) \right\} \varphi(\mathbf{R}, z) \\ &= i \left\{ \left[\frac{\Delta}{4\pi K_0} - \frac{\Delta^2}{(4\pi K_0)^3} + \dots \right] + \sigma U(\mathbf{R}, z) \right\} \varphi(\mathbf{R}, z). \end{aligned} \quad (25)$$

For readers who are not familiar with the real-space representation of the excitation error given in Eq. (25), the following relation is useful:

$$\mathbf{F} \left\{ 2\pi i K_0 \left[\sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta} - 1 \right] \varphi(\mathbf{R}, z) \right\} = 2\pi i K_0 \left[\sqrt{1 - \left(\frac{\mathbf{K}}{K_0} \right)^2} - 1 \right] \phi(\mathbf{K}, z), \quad (26)$$

where $\mathbf{F}$ indicates 2D Fourier transformation, $\mathbf{F}\{\varphi(\mathbf{R}, z)\} = \phi(\mathbf{K}, z)$, and $\mathbf{K}$ is the 2D vector in reciprocal space.

It is clear that this correction is obtained intuitively from the Ewald sphere configuration in reciprocal space (Fig. 2). However, Eq. (23) shows that such a correction is still not accurate enough. To see this clearly, we expand the second term of Eq. (23) as follows:

$$\frac{i}{4\pi K_0} \left[\frac{\Delta}{4\pi K_0} + \sigma U \right]^2 = i \left\{ \frac{\Delta^2}{(4\pi K_0)^3} + \left[\frac{\sigma^2 U^2}{4\pi K_0} + \frac{\sigma(\Delta U)}{(4\pi K_0)^2} \right] + \frac{2\sigma}{(4\pi K_0)^2} [U\Delta + (\nabla U) \cdot \nabla] \right\}, \quad (27)$$

where ∇ is the gradient operator in the x - y plane. So if term $\Delta^2/(4\pi K_0)^3$ is an important correction from the parabola to the sphere, the mixed terms in Eq. (27), $(2\sigma/(4\pi K_0)^2) [U\Delta + (\nabla U) \cdot \nabla]$, could also be significant in correction of the excitation error since both of them are proportional to λ^3 . Theoretically, the neglect of

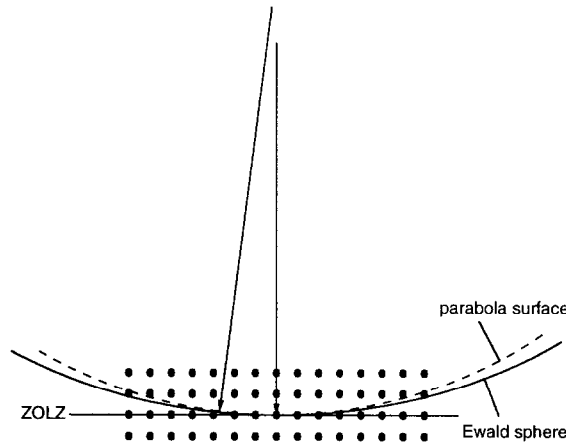


Fig. 2. Illustration of the parabolic approximation in HEED.

those mixed terms will lead to errors for the calculation of fine structures in HOLZ patterns [5, 33, 34]. In other words, **the Ewald sphere in fact is slightly modulated by the crystal potential.**

To our knowledge, it is not straightforward to implement such correction in the real-space Bloch wave method [5], although it is possible in the reciprocal space Bloch wave method (e.g., Ref. [1]). For the multislice methods, those effects have never been included. Hence to calculate HOLZ patterns accurately with the CMS, one should try to include the more accurate corrections in the procedure.

3.2. Modification of the CMS for calculating HOLZ patterns

To include the more accurate corrections in the CMS, we start from Eq. (21)

$$\Phi_{n+1}^f(\mathbf{R}) = \left(\prod_{j=1}^n e^{2\pi i \mathbf{k}_j(\mathbf{R})\varepsilon} \right) \Phi_1^f(\mathbf{R}) = e^{2\pi i \mathbf{K}_0 n \varepsilon} \left(\prod_{j=1}^n e^{2\pi i (\mathbf{k}_j - \mathbf{K}_0)\varepsilon} \right) \Phi_1^f(\mathbf{R}), \quad (28)$$

where the constant phase factor $e^{2\pi i \mathbf{K}_0 n \varepsilon}$ can be dropped in the calculation. According to Eq. (5), we make the following high-energy approximation and the multislice approximation [17, 27] for the slice-transmission operator:

$$\begin{aligned} e^{2\pi i (\mathbf{k}_j - \mathbf{K}_0)\varepsilon} &= e^{i\varepsilon[(\Delta/4\pi\mathbf{K}_0) + \sigma U_j] - (1/4\pi\mathbf{K}_0)((\Delta/4\pi\mathbf{K}_0) + \sigma U_j)^2 + \dots]} \\ &\approx e^{i\varepsilon \{ 2\pi\mathbf{K}_0 [\sqrt{1 + (\Delta/(2\pi\mathbf{K}_0)^2)} - 1] - (\sigma/(4\pi\mathbf{K}_0)^2) (\Delta U_j + U_j \Delta) + \sigma (U_j - (\sigma U_j^2/4\pi\mathbf{K}_0)) \} } \approx p(\mathbf{R}) o_j(\mathbf{R}) q_j(\mathbf{R}), \end{aligned} \quad (29)$$

where $p(\mathbf{R})$ and $q_j(\mathbf{R})$, respectively, are the pure propagator with the sphere modification and the pure phase grating with a more accurate correction:

$$\begin{aligned} p(\mathbf{R}) &= e^{2\pi i \varepsilon \mathbf{K}_0 [\sqrt{1 + (\Delta/(2\pi\mathbf{K}_0)^2)} - 1]}, \\ q_j(\mathbf{R}) &= e^{i\varepsilon \sigma (U_j - (\sigma U_j^2/4\pi\mathbf{K}_0))}, \end{aligned} \quad (30)$$

while $o_j(\mathbf{R})$ is the mixed operator with

$$o_j(\mathbf{R}) = e^{(-i\varepsilon\sigma/(4\pi\mathbf{K}_0)^2) (\Delta U_j + U_j \Delta)}. \quad (31)$$

In reciprocal space, the propagator takes the following form [31]:

$$P(\mathbf{K}) = \exp \left\{ 2\pi i \varepsilon \mathbf{K}_0 \left[\sqrt{1 - \left(\frac{\mathbf{K}}{\mathbf{K}_0} \right)^2} - 1 \right] \right\}. \quad (32)$$

So we have the modified multislice formula as follows:

$$\Phi_{n+1}^f(\mathbf{R}) = \left[\prod_{j=1}^n p(\mathbf{R}) o_j(\mathbf{R}) q_j(\mathbf{R}) \right] \Phi_1^f(\mathbf{R}). \quad (33)$$

We see that without the mixed operator, i.e., $o_j(\mathbf{R}) = 1$, Eq. (33) will return to the CMS for which the calculation can easily be performed using the FFT technique [13].

For calculating the mixed operator, we have the following two cases:

(1) For calculating the large-angle resonant HOLZ reflections, the slice thickness has to be taken so small [27] that we can rewrite Eq. (33) as

$$\Phi_{n+1}^f(\mathbf{R}) = \left[\prod_{j=1}^n o_j^2(\mathbf{R}) p(\mathbf{R}) o_j^1(\mathbf{R}) q_j(\mathbf{R}) \right] \Phi_1^f(\mathbf{R}) \quad (34)$$

and use the following approximation for the calculation:

$$\begin{aligned} o_j^1(\mathbf{R}) &= \exp\left\{-\frac{i\varepsilon\sigma}{(4\pi K_0)^2} \Delta U_j\right\} \approx 1 - \frac{i\varepsilon\sigma}{(4\pi K_0)^2} \Delta U_j, \\ o_j^2(\mathbf{R}) &= \exp\left\{-\frac{i\varepsilon\sigma}{(4\pi K_0)^2} U_j \Delta\right\} \approx 1 - \frac{i\varepsilon\sigma}{(4\pi K_0)^2} U_j \Delta. \end{aligned} \quad (35)$$

It is clear that in this case one can still use the FFT technique for the calculation, but twice more FFTs are required for each slice.

(2) For calculating ZOLZ reflections and the so-called lower-order HOLZ effects due to the surface or interface structures one can use the same procedure as that given for the large-angle resonant HOLZ reflections, as long as the slice thickness is taken so small that Eq. (35) holds. However, this may not be necessary since these reflections mainly appear in the area where the parabolic approximation (the conventional high-energy approximation) is already adequate [4] (see also the discussions in Section 5.2). So in this case we can neglect the mixed operator and therefore Eq. (33) will be reduced to the CMS for which a large slice thickness can be used for the ZOLZ calculation if the Debye–Waller factors are included for atoms [27].

In order to test the importance of the new correction for the conventional high-energy approximation, one may use the fine structures of HOLZ patterns in the CBED case where we expect the effects of the mixed operator would appear.

4. Multislice method for including back-scattering effects

4.1. High-energy approximation for calculating back-scattering

Although the rigorous solution of the Schrödinger-like equation presented in Section 2 is generally valid for all the cases ranging from LEED to HEED in solids (neglecting spin effects), the explicit calculation becomes more and more difficult with decreasing energy of the incident electron. This can be seen from the expansion of the wave-vector operator defined by Eq. (5). However, for high-energy electrons (e.g., ≥ 100 keV), as it has been discussed in Section 3, the expansion of the wave-vector operator to the second order, which is proportional to λ^3 , is sufficiently accurate for most HEED cases. We then have

$$\hat{k}_j \approx K_0 \sqrt{1 + \frac{\Delta}{(2\pi K_0)^2}} - \frac{\sigma}{2\pi(4\pi K_0)^2} (\Delta U_j + U_j \Delta) + \frac{\sigma}{2\pi} \left(U_j - \frac{\sigma U_j^2}{4\pi K_0} \right). \quad (36)$$

In fact, if an accurate correction for the excitation error is not required, even the conventional high-energy approximation will be adequate (see also, Ref. [35]), i.e.,

$$\hat{k}_j \approx K_0 + \frac{1}{2\pi} \left(\frac{\Delta}{4\pi K_0} + \sigma U_j \right). \quad (37)$$

Similarly, for HEED, the BSC operator defined by Eq. (13) can be approximated as follows:

$$\hat{B}_{j,j-1} = \frac{\hat{k}_j - \hat{k}_{j-1}}{2\hat{k}_j} \approx \frac{1}{2K_0} \left[1 - \frac{1}{2\pi K_0} \left(\frac{\Delta}{4\pi K_0} + \sigma U_j \right) \right] \left[\frac{\sigma}{2\pi} (U_j - U_{j-1}) \right] \approx \frac{\sigma}{4\pi K_0} (U_j - U_{j-1}). \quad (38)$$

We see that the magnitude of the back-scattered wave scales with the square of electron wavelength ($\sigma/K_0 \propto \lambda^2$), while that of the forward-scattered wave (for ZOLZ reflections) scales with the electron wavelength (see the expansion of the phase grating). So for HEED, the back-scattering effects are generally

much weaker than the forward-scattering effects. This suggests that we can use the so-called “single back-scattering approximation”: for the STO matrix given in Eqs. (11) and (12), we can set $\hat{s}_{j-1}^{12} = 0$ and $\hat{s}_{j-1}^{22} = e^{-2\pi i k_{j-1} \epsilon}$. In this approximation, the basic idea is that double (or multiple) back-scattering effects are negligible (being proportional to λ^4), i.e., $\hat{s}_{j-1}^{12} = 0$, and therefore $\hat{s}_{j-1}^{22} = \hat{F}_{j,j-1} e^{-2\pi i k_{j-1} \epsilon} \approx e^{-2\pi i k_{j-1} \epsilon}$.

In this way, the STO matrices are simplified as follows:

$$\hat{S}_{j-1} = \begin{pmatrix} \hat{s}_{j-1}^{11} & 0 \\ \hat{s}_{j-1}^{21} & \hat{s}_{j-1}^{22} \end{pmatrix} \quad (j = 2, 3, \dots, n, n+1) \quad (39)$$

with

$$\begin{aligned} \hat{s}_{j-1}^{11} &= (1 - \hat{B}_{j,j-1}) e^{2\pi i k_{j-1} \epsilon}, \\ \hat{s}_{j-1}^{21} &= \hat{B}_{j,j-1} e^{2\pi i k_{j-1} \epsilon}, \\ \hat{s}_{j-1}^{22} &= e^{-2\pi i k_{j-1} \epsilon}, \end{aligned} \quad (40)$$

where the wave-vector operator and the BSC operator, respectively, can be calculated from Eqs. (36) and (38).

4.2. Multislice formulations for calculating back-scattering effects

To find the multislice formulation for calculating the back-scattering wave, we substitute Eq. (39) into Eq. (15)

$$\hat{S}^n = \begin{pmatrix} \hat{s}_{11}^n & \hat{s}_{12}^n \\ \hat{s}_{21}^n & \hat{s}_{22}^n \end{pmatrix} = \begin{pmatrix} \hat{s}_n^{11} & 0 \\ \hat{s}_n^{21} & \hat{s}_n^{22} \end{pmatrix} \begin{pmatrix} \hat{s}_{n-1}^{11} & \hat{s}_{n-1}^{12} \\ \hat{s}_{n-1}^{21} & \hat{s}_{n-1}^{22} \end{pmatrix} = \begin{pmatrix} \hat{s}_n^{11} & 0 \\ \hat{s}_n^{21} & \hat{s}_n^{22} \end{pmatrix} \begin{pmatrix} \hat{s}_{n-1}^{11} & 0 \\ \hat{s}_{n-1}^{21} & \hat{s}_{n-1}^{22} \end{pmatrix} \cdots \begin{pmatrix} \hat{s}_1^{11} & 0 \\ \hat{s}_1^{21} & \hat{s}_1^{22} \end{pmatrix}, \quad (41)$$

and therefore

$$\hat{s}_{11}^n = \hat{s}_n^{11} \hat{s}_{n-1}^{11} = \hat{s}_n^{11} \hat{s}_{n-1}^{11} \cdots \hat{s}_1^{11} = \prod_{j=2}^{n+1} (1 - \hat{B}_{j,j-1}) e^{2\pi i k_{j-1} \epsilon}, \quad (42)$$

$$\hat{s}_{12}^n = 0, \quad (43)$$

$$\hat{s}_{22}^n = \hat{s}_n^{22} \hat{s}_{n-1}^{22} = \hat{s}_n^{22} \hat{s}_{n-1}^{22} \cdots \hat{s}_1^{22} = \prod_{j=2}^{n+1} e^{-2\pi i k_{j-1} \epsilon}, \quad (44)$$

and

$$\begin{aligned} \hat{s}_{21}^n &= \hat{s}_n^{21} \hat{s}_{n-1}^{11} + \hat{s}_n^{22} \hat{s}_{n-1}^{21} = \hat{s}_n^{21} \hat{s}_{n-1}^{11} + \hat{s}_n^{22} (\hat{s}_{n-1}^{21} \hat{s}_{n-2}^{11} + \hat{s}_{n-1}^{22} \hat{s}_{n-2}^{21}) \cdots \\ &= \sum_{m=1}^n \left(\prod_{j=m+1}^n \hat{s}_j^{22} \right) \hat{s}_m^{21} \hat{s}_{11}^{m-1}. \end{aligned} \quad (45)$$

In Eq. (45) we define $\hat{s}_{11}^0 = 1$ and $\hat{s}_{n+1}^{22} = 1$. On the other hand, since $(\hat{s}_{22}^n)^{-1} \hat{s}_{22}^n = \hat{s}_{22}^n (\hat{s}_{22}^n)^{-1} = 1$, from Eq. (44) we have

$$(\hat{s}_{22}^n)^{-1} = \prod_{j=n}^1 e^{2\pi i k_j \epsilon}. \quad (46)$$

So we finally obtain the forward-scattered wave [Eq. (20)]

$$\Phi_{n+1}^f = \hat{s}_{11}^n \Phi_1^f = \left[\prod_{j=2}^{n+1} (1 - \hat{B}_{j,j-1}) e^{2\pi i k_{j-1} \epsilon} \right] \Phi_1^f, \quad (47)$$

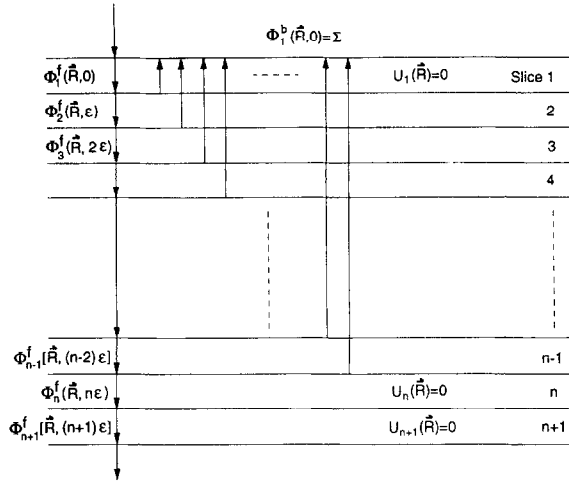


Fig. 3. Schematic illustration of Eq. (47) for forward-scattering and Eq. (48) for back-scattering.

and the back-scattered wave [Eq. (19)]

$$\begin{aligned}\Phi_1^b &= -(\hat{s}_{22}^n)^{-1} \hat{s}_{21}^n \Phi_1^f = -(\hat{s}_{22}^n)^{-1} \left[\sum_{m=1}^n \left(\prod_{j=m+1}^n \hat{s}_j^{22} \right) \hat{s}_m^{21} \hat{s}_{11}^{m-1} \right] \Phi_1^f \\ &= - \sum_{m=1}^n \left((\hat{s}_{22}^n)^{-1} \prod_{j=m+1}^n \hat{s}_j^{22} \right) \hat{s}_m^{21} (\hat{s}_{11}^{m-1} \Phi_1^f) = - \sum_{m=1}^n \left(\prod_{j=m}^1 e^{2\pi i k_j \epsilon} \right) (\hat{B}_{m+1, m} e^{2\pi i k_m \epsilon}) \Phi_m^f\end{aligned}\quad (48)$$

since we have Eqs. (40) and (46) and therefore we have

$$\begin{aligned}(\hat{s}_{22}^n)^{-1} \prod_{j=m+1}^n \hat{s}_j^{22} &= \left(\prod_{j=n}^1 e^{2\pi i k_j \epsilon} \right) \left(\prod_{j=m+1}^n e^{-2\pi i k_j \epsilon} \right) \\ &= e^{2\pi i k_1 \epsilon} \dots e^{2\pi i k_m \epsilon} (e^{2\pi i k_{m+1} \epsilon} \dots e^{2\pi i k_n \epsilon} e^{-2\pi i k_n \epsilon} \dots e^{-2\pi i k_{m+1} \epsilon}) = \left(\prod_{j=m}^1 e^{2\pi i k_j \epsilon} \right).\end{aligned}\quad (49)$$

The whole procedure is schematically illustrated in Fig. 3. For HEED, the wave-vector operator and the BSC operator, respectively, can be approximated by Eqs. (36) and (38) and therefore the techniques which have been established for the CMS can still be used for performing the calculations.

5. Discussions

5.1. Multislice theory

The CMS has been a powerful method in computational electron microscopy, but its applicability to some HEED cases, such as HOLZ effects and large-angle inclined illumination, has never been convincing [36, 10, 37]. The basic reason for this is that it is based on the reduced Schrödinger equation [Eq. (24)] [37, 17]. However, we have seen that the multislice theory can be developed from the complete Schrödinger equation for elastic electron scattering. This development allows the theory to include more accurate effects if necessary. Hence the multislice theory can in principle be as accurate as the Bloch wave theory (but this theory can not include surface effects) and the Green function theory in all cases of elastic electron scattering.

It is interesting to notice that in RHEED calculations, there is another type of multislice method, i.e., the so-called *R*-matrix multislice theory (e.g., Ref. [38]). In principle, the *R*-matrix multislice theory and the operator multislice theory presented in this work should be equivalent since both are based on the complete Schrödinger equation. However, the operator multislice theory is a further development of the Cowley–Moodie multislice theory, in which the physico-optical concepts of phase grating and propagator play the key rules for numerical calculations. But the *R*-matrix multislice theory does not have such concepts and it is essentially a reciprocal-space matrix method. On the other hand, in the present work, we are mainly interested in the transmitted wave and back-scattering effects are included only based on the “single back-scattering approximation”, although Eqs. (19) and (20) are accurately valid for all scattering effects.

5.2. Forward-scattering approximation

The accurate simulation of HOLZ patterns in CBED case is important for the accurate measurement of local lattice parameters [39]. Currently, the real-space Bloch wave calculations of HOLZ patterns are based on the conventional high-energy approximation for forward-scattering including the correction from the parabolic to the spherical surface [5]. However, it is shown by Eq. (23) that if the spherical surface correction is significant, some other corrections should also be included. The importance of such corrections as function of the electron wavelength can numerically be checked from Eq. (23). Theoretically, such corrections are expected to be significant in the following two cases:

- (i) low or medium energy electron diffraction,
- (ii) in cooled crystals where many large-angle HOLZ rings can appear [40].

It has been shown how the CMS takes HOLZ effects into account [27]. The same analysis can be made for the new procedure presented in Section 3.2 and this leads to a new expression for the wavefield in case of ZOLZ interaction (without the potential variation along the *z*-axis):

$$\begin{aligned}
 \Phi_{n+1}^{\text{ZOLZ}}(\mathbf{R}) &= \exp \left\{ 2\pi i n \varepsilon \sum_{j=1}^n \frac{(\hat{k}_j - K_0)}{n} \right\} \Phi_1^t(\mathbf{R}) \\
 &= \exp \left\{ 2\pi i n \varepsilon K_0 \sum_{j=1}^n \frac{1}{n} \left(\sqrt{1 + \frac{\Delta}{(2\pi K_0)^2} + \frac{\sigma}{\pi K_0} U_j} - 1 \right) \right\} \Phi_1^t(\mathbf{R}) \\
 &\approx \exp \left\{ 2\pi i n \varepsilon K_0 \left(\sqrt{1 + \frac{\Delta}{(2\pi K_0)^2} + \frac{\sigma}{\pi K_0} \bar{U}} - 1 \right) \right\} \Phi_1^t(\mathbf{R})
 \end{aligned} \tag{50}$$

where $\bar{U}(\mathbf{R})$ is the potential averaged over whole the crystal. However, for HEED Eq. (50) will not lead to significant corrections for the conventional definition of ZOLZ effects if the total thickness of the solid foil is around a few ten nanometers, since (i) the correction terms are proportional to λ^3 and (ii) ZOLZ reflections are located in the area where the parabolic approximation is already sufficiently accurate (Fig. 2).

5.3. Back-scattering and large beam tilt

From the point of view of the Ewald sphere configuration (Fig. 4), the back-scattering reflections are in fact HOLZ effects due to the potential variation along the beam direction. The appearance of back-scattering effects depends on the incident beam direction, the crystal structure, the temperature factors of atoms, the crystal thickness and the electron wavelength as well. However, to include back-scattering effects in the formulations presented in Section 4, we first of all have to clarify the concept of back-scattering. In the theoretical scheme, the back-scattering is defined with respect to the slice normal (the *z*-axis) (Fig. 5a), not to the incident beam direction (Fig. 5b). For HEED, those back-scattering effects defined with respect to the

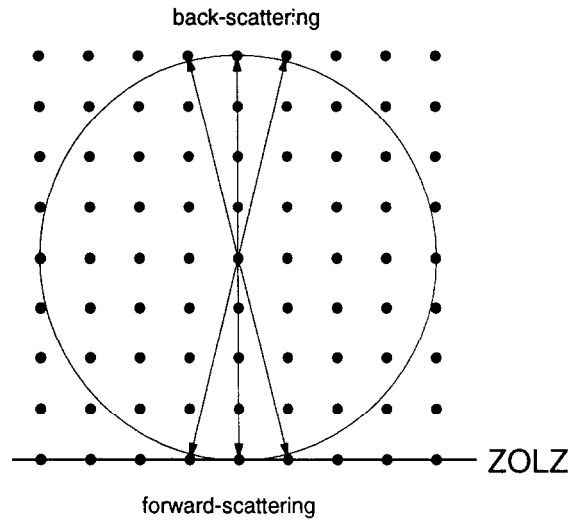


Fig. 4. Ewald sphere configuration for electron scattering in crystals.

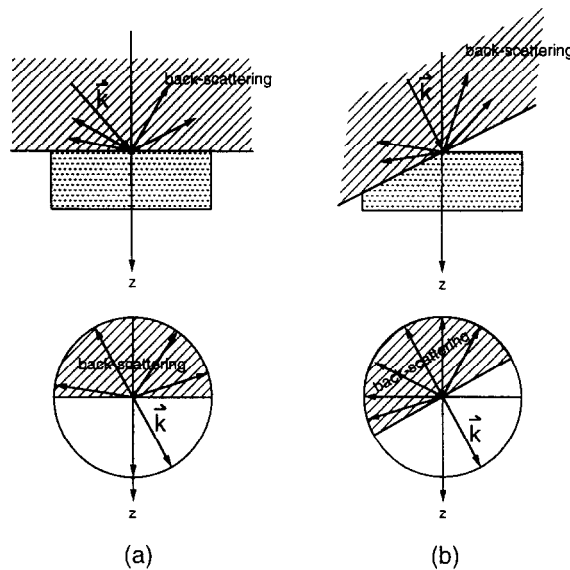


Fig. 5. Definitions of back-scattering effects with respect to (a) the slice normal and (b) the incident beam direction.

incident beam can hardly occur, because of the weak scattering factors of atoms for large-angle scattering. For the multislice theory, in principle, one can always take the incident beam direction as the slice normal, so that no beam tilt effects have to be included and also back-scattering effects can reasonably be neglected. In practice, however, it is not always possible or convenient to do so, so that beam tilt and back-scattering effects have to be included in the multislice method. For example, in order to calculate the transmitted wave from a non-orthogonal crystal where the incident beam is along the zone axis (Fig. 6a) and that from an orthogonal crystal where the incident beam is largely tilted away from the surface normal (Fig. 6b), it is

对于多层理论，原则上可以将入射光束方向作为切片正线，这样就不需要考虑光束倾斜效应，也可以合理地忽略后向散射效应。

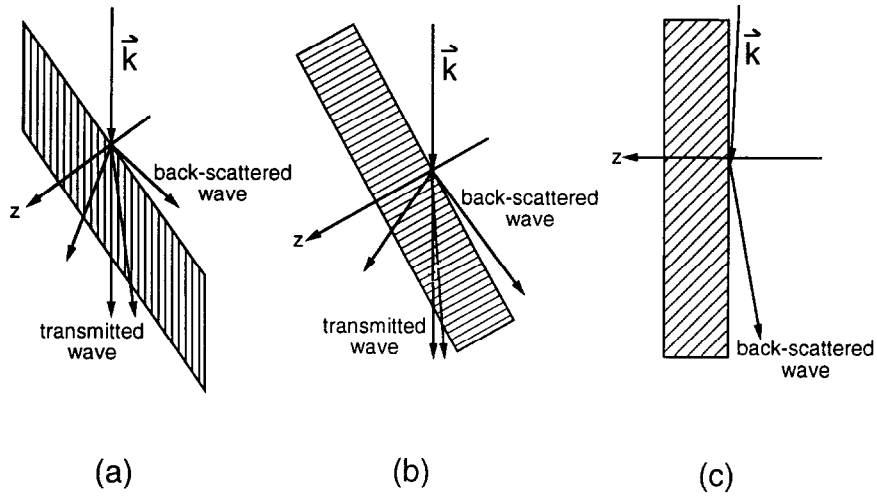


Fig. 6. Illustration of some possible back-scattering effects: (a) electron diffraction in non-orthogonal crystals; (b) electron diffraction in orthogonal crystals; (c) RHEED case.

impossible to calculate the phase gratings if the slice normal is taken parallel to the incident beam direction. In these cases, the formula, Eq. (47) may be useful since it takes into account the beam tilt as well as back-scattering effects.

It is known that the CMS is not valid for a large beam tilt [2, 10, 41]. From the above discussion we see that this is because the CMS does not include back-scattering effects even in case of large beam tilt. An extreme example to show the importance of back-scattering effects for large beam tilt is RHEED, where the incident beam is tilted so as to be almost perpendicular to the surface normal (Fig. 6c). However, it should be pointed out that for RHEED case, where the back-scattering effects are dominant, Eqs. (47) and (48) may not be sufficient since they are based on the “single back-scattering approximation”.

5.4. Calculations of the new multislice formulations

To include the correction operator Eq. (31) in the multislice method for calculating the large-angle resonant HOLZ reflections, one needs twice the number of FFTs as compared to a CMS procedure and therefore the total computing time will approximately be doubled. It should be stressed that for the calculation of HOLZ reflections with the multislice method, the slice thickness has to be made so small that the multislice potential distribution approaches the real potential distribution in the solid, in order to guarantee the convergence of the calculations [27].

For inclined illumination, the beam tilt effect can be either included in incident wave function $\Phi_1^i(\mathbf{R})$ by a plane wave factor $e^{2\pi i \mathbf{K}_u \cdot \mathbf{R}}$, or transferred into the multislice formulations [42]. The latter is convenient for performing multislice calculations using the FFT technique. The transfer is done by changing all the operations of $\Delta \Phi(\mathbf{R})$ into $[\Delta + 4\pi i \mathbf{K}_u \cdot \nabla + (2\pi i \mathbf{K}_u)^2] \Phi(\mathbf{R})$ in real space, or correspondingly, changing $(2\pi i \mathbf{K})^2 \phi(\mathbf{K})$ into $(2\pi i)^2 (\mathbf{K} + \mathbf{K}_u)^2 \phi(\mathbf{K})$ in reciprocal space where $\phi(\mathbf{K})$ is the Fourier transform of the wavefunction $\Phi(\mathbf{R})$ and ∇ is the gradient operator in the x - y plane.

From Eq. (47) we see that the calculation of the transmitted wave which includes the back-scattering effects can easily be done by the multislice procedure presented in Section 4.2, provided the phase gratings are corrected by the factors $(1 - \hat{B}_{i,j-1}) \approx [1 - \sigma/4\pi K_0 (U_j - U_{j-1})] \approx \exp\{-\sigma/4\pi K_0 (U_j - U_{j-1})\}$, so as to be $\exp\{i\sigma\epsilon[U_j + (i\lambda/4\pi\epsilon)(U_j - U_{j-1})]\}$. However, the calculation of the back-scattered wave itself will be

much more time consuming unless a large amount of computer memory can be used for saving the transmitted functions of all the slices. Fortunately, in practical cases of HEED, only the influence of the back-scattering effects on the transmitted wave are interesting. It should be noted that the back-scattering appears as a kind of absorption effect in the transmitted wave so that the conventional test of normalisation for the transmitted beams will not be accurate.

6. Conclusions

The Schrödinger equation for elastic electron scattering in solids can be solved rigorously within the framework of the multislice scheme. In order to do so, the concept of the slice-transmission-operator (STO) matrix and the transfer matrix technique are very useful since they lead to a more general and accurate multislice theory. This new approach still preserves the important computational advantages of the conventional multislice method over the Bloch wave method and therefore is generally valid for any type of solid structures (including surfaces).

The new approach leads to accurate corrections for the conventional high-energy approximation which serves as the basis of many existing dynamical theories for elastic forward-scattering. Using the so-called “single back-scattering approximation”, an improved multislice formula is obtained to calculate the transmitted wave. This formula takes back-scattering effects into account and may therefore be used for large beam tilt. It is shown that most of the numerical techniques of the conventional multislice procedure, including the FFT technique, can still be used for performing the new multislice procedure.

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